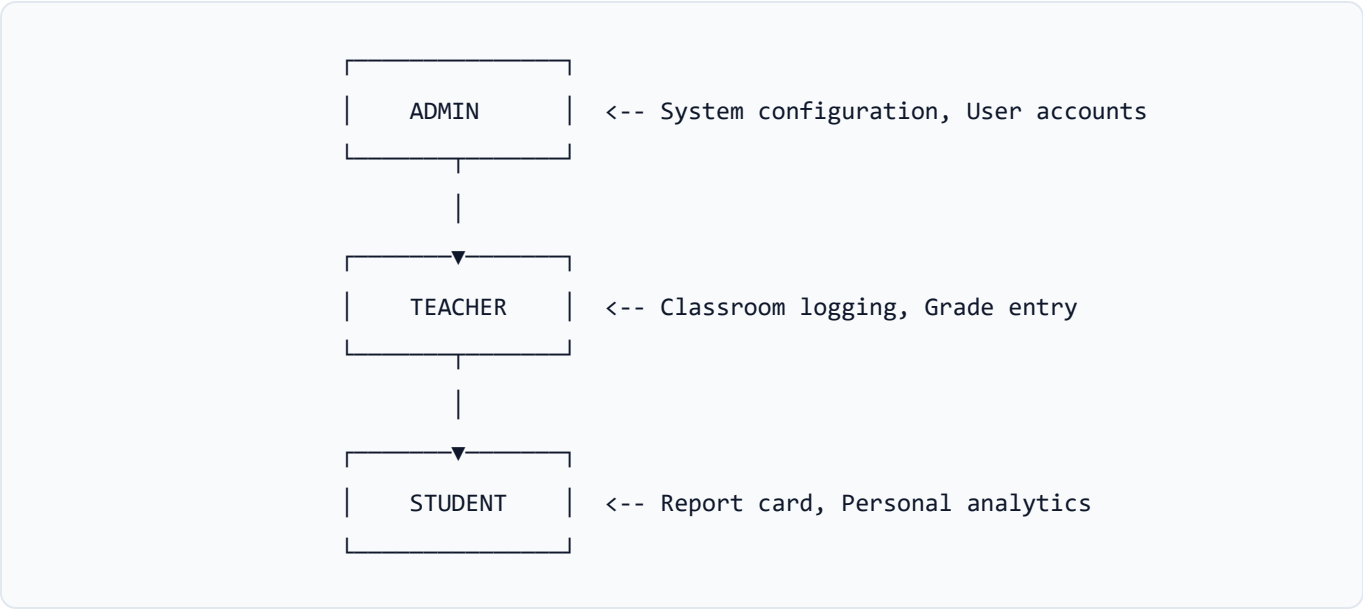


User Roles and Permissions Matrix - Student Performance Analytics Portal

This document clarifies the access levels, credentials, and actions permitted for each of the three user roles defined in the Student Performance Analytics Portal system.

Roles Overview



Detailed Permissions Matrix

Feature / Action	Admin	Teacher	Student	Details / Notes
Login / Logout	Yes	Yes	Yes	All users authenticate using JWT
View Profile Settings	Yes	Yes	Yes	Can reset own password
Create/Edit User Logins	Yes	No	No	Only Admin can provision accounts

Feature / Action	Admin	Teacher	Student	Details / Notes
Register Student Profiles	Yes	No	No	Link User to demographic Student schema
Assign Course to Teacher	Yes	No	No	Map course code to teacher User ID
Log Grades	Yes	Yes	No	Teacher logs marks for assigned courses only
Edit/Delete Grades	Yes	Yes	No	Allow corrections before term lock
Log Daily Attendance	Yes	Yes	No	Teacher marks student attendance
View Global Institution KPIs	Yes	No	No	Access global school performance dashboard
View Class-Level Analytics	Yes	Yes	No	Average GPA, distribution charts by class
View Student Profile Sheet	Yes	Yes	Self Only	Teachers see students they teach
View Personal Grade Card	Yes	Yes	Self Only	Student sees own score cards
View Personal Attendance History	Yes	Yes	Self Only	Student reviews presence percentages

Security Enforcement Rules

1. Role-Based Token Verification:

- The Express backend reads the JWT, verifies the signature, and matches the embedded `role` value against route permissions using `authorize('admin', 'teacher')` .

2. Data Scope Isolation:

- For student access requests, the backend controller confirms that the queried student ID matches the authenticated user's ID (`req.user.id === student.userId`), preventing cross-tenant access.